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C

For X:  $R = \frac{\rho L}{A_x}$

$$R = \frac{\rho L}{V / L} = \frac{\rho L^2}{V} \text{----- (1)}$$

For Y:  $R_Y = \frac{\rho(2L)}{V / 2L} = \frac{4\rho L^2}{V} \text{----- (2)}$

Same volume  $V$ . Hence  $A_Y = V / 2L$ . Also, same metal hence,  $\rho$  is the same.  
Therefore,

$$R_Y = 4R$$

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A

$I = E / \text{Total resistance}$